

D491

Introduction to Analytics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional grocery chain with 42 stores uses a model that recommends how much of each item to order next week to minimize waste, based on forecast demand and shelf life. The model runs on a cloud platform the chain adopted in 2022. Which type of analytics is this?

- A. Descriptive.
- B. Diagnostic.
- C. Prescriptive.
- D. Predictive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A manager asks, “Why did product returns jump 30% in March?” Which type of analytics answers this question?

- A. Predictive.
- B. Prescriptive.
- C. Descriptive.
- D. Diagnostic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Seven employees on a small sales team earned these bonuses last year: \$2,000, \$2,500, \$3,000, \$3,000, \$3,500, \$4,000, and \$40,000. The team sells industrial pumps across four states. Which value best represents a typical bonus?

- A. \$40,000
- B. \$3,000 (the median)
- C. \$38,000 (the range)
- D. About \$8,286 (the mean)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A data set of home prices has a mean of \$410,000 and a median of \$330,000. What does this most likely indicate?

- A.** The data is skewed left.
- B.** The mean is always the better summary of home prices.
- C.** A few very expensive homes pull the mean above the median (right skew).
- D.** Most homes cost more than \$410,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An analyst at a city parks department finds that months with higher ice cream sales also have more swimming-pool injuries ($r = 0.82$). The analysis used five years of data from 11 pools. What is the best conclusion?

- A.** Ice cream sales cause pool injuries.
- B.** The two are strongly related, probably through a shared cause such as hot weather; this does not show that ice cream causes injuries.
- C.** Because r is above 0.8, the link is proven to be causal.
- D.** The relationship is weak because only 11 pools were studied.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A hotel survey asks guests to rate their stay as poor, fair, good, or excellent. The survey was emailed to 3,000 guests in June. What level of measurement is the rating?

- A.** Ordinal.
- B.** Ratio.
- C.** Nominal.
- D.** Interval.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which variable is measured at the ratio level?

- A.** Customer ZIP code.
- B.** Delivery time in minutes.
- C.** Temperature in degrees Fahrenheit.
- D.** Satisfaction rank from 1 to 5.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A media company with 2 million subscribers needs someone to build and maintain automated pipelines that move raw viewing logs into its data warehouse every night. The company's headquarters moved to Denver last year. Which role fits best?

- A. Data engineer.
- B. Data scientist.
- C. Data analyst.
- D. Any analytics role, since every role owns pipelines.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

To estimate how satisfied all 20,000 customers of an internet provider are, an analyst surveys only the 400 customers who called the support line last week. The survey took 6 minutes on average. What is the main problem?

- A. The sample is biased; customers who called support are not representative of all customers.
- B. The 6-minute survey was too long.
- C. The analyst should survey all 20,000 because sampling never works.
- D. 400 is always too few for a population of 20,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

On a warehouse productivity test, workers average 80 picks per hour with a standard deviation of 5. One worker, who joined in April, scored 92. What is her z-score?

- A. -2.4
- B. 18.4
- C. 12
- D. 2.4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Test scores have a mean of 70 and a standard deviation of 8. What raw score has a z-score of -1.5 ?

- A. 82
- B. 68.5
- C. 58
- D. 62

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A logistics firm with a new analytics team of six wants to reduce late deliveries. The team lead is eager to try a neural network she read about. Under CRISP-DM, what should the team do first?

- A. Start cleaning all available delivery data.
- B. Build the neural network to see what it finds.
- C. Deploy a dashboard of late deliveries.
- D. Define the business problem and success criteria with stakeholders.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A regional grocery chain ~~with 42 stores~~ uses a model that recommends how much of each item to order next week to minimize waste, based on forecast demand and shelf life. ~~The model runs on a cloud platform the chain adopted in 2022.~~ Which type of analytics is this?

Target word: analytics type **Decides:** It recommends an action (how much to order).

A	Confusion set	Descriptive summarizes what already happened.
B	Confusion set	Diagnostic explains why something happened.
C	Correct	It says what to do, using a forecast as input.
D	Adjacent category	The demand forecast is predictive, but the output here is a recommended action.

2. Answer D Concept 1 • transfer

A manager asks, “Why did product returns jump 30% in March?” Which type of analytics answers this question?

Target word: type from the question — reverse **Decides:** The question asks why.

A	Inverted operation	Looks forward instead of back.
B	Confusion set	Recommends actions; it does not explain the past.
C	Adjacent category	Descriptive would report the 30% jump, not explain it.
D	Correct	Explaining causes of a past change is diagnostic.

3. Answer B Concept 2

Seven employees on a small sales team earned these bonuses last year: \$2,000, \$2,500, \$3,000, \$3,000, \$3,500, \$4,000, and \$40,000. ~~The team sells industrial pumps across four states.~~ Which value best represents a typical bonus?

Target word: typical value **Decides:** One extreme value (\$40,000) skews the data.

A	Wrong stem number	The outlier itself.
B	Correct	The middle of the seven sorted values; unaffected by the \$40,000 outlier.
C	Adjacent category	The range measures spread, not a typical value.
D	Confusion set	\$58,000 / 7 is pulled up by one outlier; six of seven bonuses are below \$4,100.

4. Answer C Concept 2 • transfer

A data set of home prices has a mean of \$410,000 and a median of \$330,000. What does this most likely indicate?

Target word: infer shape from mean vs. median — reverse **Decides:** Mean well above median.

A	Inverted operation	Left skew would put the mean below the median.
B	Absolute language	With skew, the median is usually the better typical value.
C	Correct	High outliers raise the mean more than the median.
D	Confusion set	Half the homes are below \$330,000.

5. Answer B Concept 3

An analyst at a city parks department finds that months with higher ice cream sales also have more swimming-pool injuries ($r = 0.82$). ~~The analysis used five years of data from 11 pools.~~ What is the best conclusion?

Target word: what the correlation supports **Decides:** Correlation alone; hot weather likely drives both.

A	Confusion set	Confuses correlation with causation.
B	Correct	A strong correlation with a likely third variable does not support causation.
C	Absolute language	No r value proves causation.
D	Camouflage	The pool count does not make $r = 0.82$ weak.

6. Answer A Concept 4

A hotel survey asks guests to rate their stay as poor, fair, good, or excellent. ~~The survey was emailed to 3,000 guests in June.~~ What level of measurement is the rating?

Target word: level of measurement **Decides:** Ordered categories with no fixed gap.

A	Correct	The categories have an order, but the gaps between them are not equal.
B	Inverted operation	The highest level; needs equal gaps and a true zero.
C	Confusion set	Nominal labels have no order; these do.
D	Adjacent category	Interval needs equal gaps between values.

7. Answer B Concept 4 • transfer

Which variable is measured at the ratio level?

Target word: find the ratio variable — reverse **Decides:** A true zero meaning “none.”

A	Confusion set	ZIP codes are nominal labels, even though they are digits.
B	Correct	Zero minutes means no time; 20 minutes is twice 10.
C	Adjacent category	Interval: 0°F is not “no temperature.”
D	Confusion set	Ordinal: ordered, uneven gaps.

8. Answer A Concept 5

A media company ~~with 2 million subscribers~~ needs someone to build and maintain automated pipelines that move raw viewing logs into its data warehouse every night. ~~The company's headquarters moved to Denver last year.~~ Which role fits best?

Target word: role **Decides:** Building and maintaining pipelines.

A	Correct	Data engineers build pipelines and storage.
B	Adjacent category	Data scientists build models on data the pipelines deliver.
C	Confusion set	Analysts query and report; they rarely own nightly pipelines.
D	Absolute language	Roles have distinct focuses.

9. Answer A Concept 6

To estimate how satisfied all 20,000 customers of an internet provider are, an analyst surveys only the 400 customers who called the support line last week. ~~The survey took 6 minutes on average.~~ What is the main problem?

Target word: flaw in the sample **Decides:** Only support callers were sampled.

A	Correct	People with problems are overrepresented.
B	Camouflage	Survey length does not change who was chosen.
C	Adjacent category	Random sampling works; biased sampling does not.
D	Absolute language	A random sample of 400 can be adequate; the selection method is the issue.

10. Answer D Concept 7

On a warehouse productivity test, workers average 80 picks per hour with a standard deviation of 5. One worker, ~~who joined in April,~~ scored 92. What is her z-score?

Target word: z-score **Decides:** $(92 - 80) / 5$.

A	Inverted operation	Subtracts in the wrong order; 92 is above the mean.
B	Wrong stem number	92 / 5 skips subtracting the mean.
C	Dropped element	The raw distance; it skips dividing by the SD.
D	Correct	$12 / 5 = 2.4$ standard deviations above the mean.

11. Answer C Concept 7 • transfer

Test scores have a mean of 70 and a standard deviation of 8. What raw score has a z-score of -1.5 ?

Target word: solve for the raw value **Decides:** Value = mean + $z \times$ SD.

A	Inverted operation	Adds 12 instead of subtracting it.
B	Dropped element	Subtracts 1.5 without multiplying by the SD.
C	Correct	$70 + (-1.5 \times 8) = 70 - 12 = 58$.
D	Wrong stem number	Subtracts one SD (8) instead of 1.5 SDs.

12. Answer D Concept 8

A logistics firm ~~with a new analytics team of six~~ wants to reduce late deliveries. ~~The team lead is eager to try a neural network she read about.~~ Under CRISP-DM, what should the team do first?

Target word: first step **Decides:** CRISP-DM starts with business understanding.

A	Dropped element	Skips defining the problem, so it is unclear which data matters.
B	Camouflage	Answers the lead's preference; modeling comes later.
C	Inverted operation	Deployment is the last phase.
D	Correct	Business understanding comes first.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Four types of analytics	Descriptive = what happened; diagnostic = why it happened; predictive = what will happen; prescriptive = what to do about it.	Calling any forecast or model “prescriptive,” or any report “diagnostic.”	The question word: what, why, what will, what should.
2	Mean vs. median	The median resists outliers; when data is skewed, the mean is pulled toward the long tail.	Reporting the mean as “typical” when an extreme value is present.	One value far from the rest, or mean and median far apart.
3	Correlation is not causation	Two variables moving together can share a third cause; only a controlled design supports a causal claim.	Treating a strong r as proof that one variable causes the other.	A high correlation and a causal-sounding answer choice.
4	Levels of measurement	Nominal = labels; ordinal = ordered labels with uneven gaps; interval = equal gaps, no true zero; ratio = equal gaps and a true zero.	Treating rating scales as interval or temperature as ratio.	Ranked categories, or whether zero means “none.”
5	Analytics roles	Data engineers build and maintain pipelines and storage; data analysts turn data into reports and answers; data scientists build statistical and machine learning models.	Assuming any analytics role owns every task.	Pipelines and warehouses vs. dashboards vs. models.
6	Sampling bias	A sample supports conclusions about a population only if it represents that population; size cannot fix a biased selection method.	Thinking a big enough sample is always valid.	The sample was drawn from a special subgroup.
7	Standard deviation and z-scores	$z = (\text{value} - \text{mean}) / \text{standard deviation}$; it counts how many standard deviations a value sits from the mean.	Forgetting to subtract the mean or to divide by the SD.	A mean, an SD, and one value.
8	Start with the business question	In CRISP-DM, business understanding comes first, then data understanding, preparation, modeling, evaluation, and deployment.	Jumping to a favorite model or cleaning data before defining the problem.	“What should the team do first?”

Course scope (public description): Analytics is the creative use of data and statistical modeling to tell a compelling story that not only drives strategic action, but also results in business value. Introduction to Analytics examines data analytics as a discipline and the various roles and functions within the field. You will expand your knowledge about what analytics is and develop a basic understanding of statistics, analysis, problem solving, and programming concepts.